

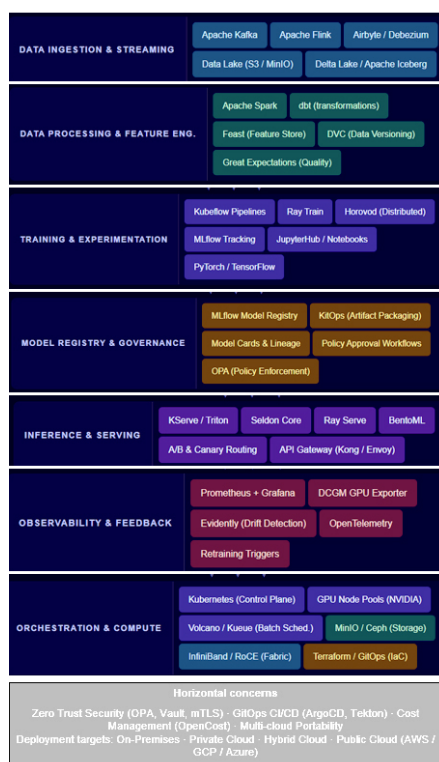


Figure 2: A layered, cloud-agnostic reference design for an enterprise-grade AI/ML infrastructure platform

The build vs buy vs compose decision

Enterprise architects face a structural choice in AI infrastructure: build custom platforms from raw open source components, adopt managed cloud services (Amazon SageMaker, Google Vertex AI, Azure ML), or compose a best-of-breed stack from a combination of both. The correct answer is almost always a composition strategy, but the specific composition depends on three factors that every organisation must honestly assess.

The first factor is engineering maturity. Deploying and operating a production-grade open source ML platform requires significant Kubernetes expertise, MLOps engineering capability, and operational discipline. Organisations that lack this capacity will find that the theoretical cost advantages of open source evaporate in operational overhead and reliability incidents. The second factor is data sovereignty. Organisations

in regulated industries — banking, healthcare, defence — often cannot place training data or model artifacts in managed cloud services, making self-hosted or co-located infrastructure architecturally mandatory regardless of cost. The third factor is workload predictability. Highly variable AI workloads (burst training, seasonal inference patterns) favour cloud elasticity; sustained high-utilisation workloads (continuously serving large LLMs at enterprise scale) favour on-premises economics.

The hybrid approach — using managed cloud for elasticity and experimentation while maintaining self-hosted infrastructure for sustained production workloads and sensitive data — has become the dominant pattern in sophisticated enterprise AI deployments. Designing this hybrid coherently, with consistent governance and observability across both environments, is one of the most important architectural challenges in the field today.

Cost architecture: The hidden design dimension

Infrastructure cost is rarely treated as a first-class architectural concern in AI platform design, and this omission is consistently expensive. GPU compute costs, data storage and egress, and the operational overhead of platform teams can collectively make AI unit economics untenable even when individual models are technically successful. Cost architecture — designing for cost visibility, allocation, and optimisation from the outset — is as important as performance architecture.

Practical cost architecture involves several interlocking practices. GPU cluster utilisation must be actively managed; idle reserved instances are extraordinarily wasteful at GPU prices. Spot instance strategies for non-time-sensitive training workloads can reduce compute costs by 60–80% but require fault-tolerant training pipelines that can checkpoint and

resume. Data tiering — automatically moving cold training data and model artifacts to cheaper storage classes — can dramatically reduce storage costs without compromising accessibility. And team-level cost visibility, implemented through namespace-level resource tagging and show back reporting, creates the accountability mechanisms that drive continuous efficiency improvement.

Organisational readiness: The infrastructure is only half the problem

Every infrastructure investment must be matched by corresponding investment in the human systems that operate and evolve it. The current state of enterprise AI talent is a matter of concern — only 14% of enterprise leaders report having the talent necessary to meet their AI infrastructure goals, according to recent survey data. This is not primarily a recruiting problem — it is an organisational design problem.

The most effective organisational model for enterprise AI infrastructure is a dedicated platform engineering team — analogous to a site reliability engineering function — the explicit charter of which is to build and operate the ML platform as an internal product. This team owns the infrastructure, defines standards, and creates the self-service tooling that allows data science teams to move fast without each team reinventing the wheel or creating technical debt. Organisations that have invested in this model consistently outperform those that treat AI infrastructure as a shared responsibility of existing IT operations teams.

Equally important is the feedback loop between platform engineers and the data scientists and ML engineers who use the platform. The costliest infrastructure mistakes — inadequate storage for large model checkpoints, training cluster network configurations that bottleneck distributed jobs, observability gaps that make drift invisible — are often diagnosed in hindsight by practitioners who could